

Abdelrahman Eissa

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EDUCATION

York University – Lassonde School of Engineering

B.Eng., Spec. Hons. Software Engineering GPA: 8.96 / 9.0

Toronto, ON

Expected Graduation, 2028

EXPERIENCE

Research Assistant – Open Modulation Classification

May 2026 – Present

York University

Toronto, ON

- Architected and coded an incremental learning pipeline to classify dynamic wireless signals, successfully converting ambiguous open-set data into structured closed-set classification tasks.
- Built a filtering module (Moving Intersection Algorithm) and integrated DBSCAN clustering algorithms to dynamically categorize unrecognized data streams into multi-modal distributions without prior labeling.
- Optimized model training loops with Batch-All Triplet Loss and sample replay techniques to seamlessly expand output capacity without suffering from catastrophic data forgetting.
- Operated in a fast-paced research environment, continuously delivering, debugging, and integrating advanced algorithmic features on a bi-weekly sprint schedule.

Software Engineer Intern

June 2025 – Aug. 2025

BWell (Health-Tech Startup)

- Designed and implemented the core diagnostic module of an agentic workflow that automatically triages patients.
- Architecture inspired by Microsoft's *AI Clinician* paper achieving 100% accuracy on in-house and physician-curated test cases.
- Deployed pipeline using Python, FastAPI, and LLM orchestration tools.

Peer Mentor

June 2025 – April 2026

York University

Toronto, ON

- Provided academic mentorship for MATH 1028 and MATH 1025, supporting engineering students with complex course material and academic transitions.

COMPETITIONS

York Engineering Competition (YEC) | *Programming*

2024

- Secured 3rd place in Programming for an optimization tool designed to improve Google Maps transit routing.
- Calculated optimal arrival times to significantly minimize waiting intervals between transit transfers.

PROJECTS

Competitor Intelligence & Financial Scraper | *Python, LLM, SQL, Cron, Webhooks*

2025

- Engineered an automated data extraction pipeline leveraging LLM-assisted HTML parsing and scheduled Cron workers for multi-source competitor price tracking.
- Architected user authentication workflows and database tracking backend to normalize scraped product data and store historical metrics.
- Built real-time email notification modules for live price shifts, delivering a production-ready system actively deployed for commercial e-commerce workflows.

Multilayer Perceptron (MLP) from Scratch | *C++*

2025

- Architected a Multilayer Perceptron (MLP) from the ground up using C++ without external ML libraries.
- Implemented fundamental neural network components including the forward pass and backpropagation algorithms.
- Validated the model against the MNIST dataset, achieving a 96% classification accuracy.

Automatic Assignment Marker | *Java, Google Vision API, Spring Boot*

2024

- Engineered an automated marking system using Java and Spring Boot to streamline the grading process.
- Integrated the platform with Google Classroom API to maintain existing teacher workflows.
- Incorporated AI to interpret student intent during grading, increasing marking accuracy to 98%.
- Reduced student feedback latency by eliminating manual grading bottlenecks.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL, HTML/CSS

Libraries/Tools: Google Vision API, Flask, FastAPI, React, Git, Docker, LLM orchestration, Machine Learning, Web Scraping